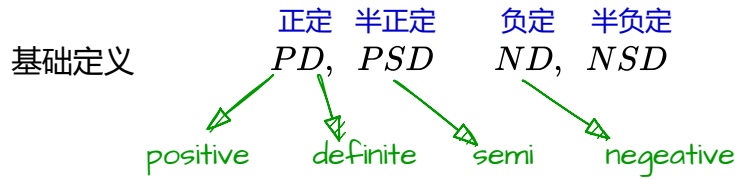


Advanced 控制理论 nonlinear 非线性

Lyapunov Direct Method



对于函数 $V: D \rightarrow \mathbb{R}$

$PD: (1) V(0) = 0$

$V(x) > 0, \forall x \in D - \{0\}$

$ND: (1) V(0) = 0$

$V(x) < 0, \forall x \in D - \{0\}$

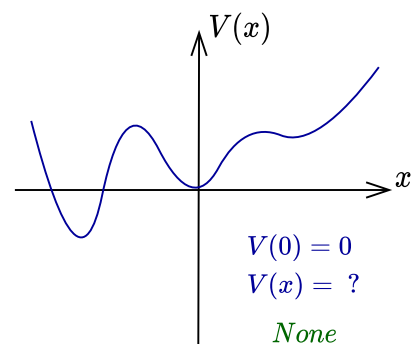
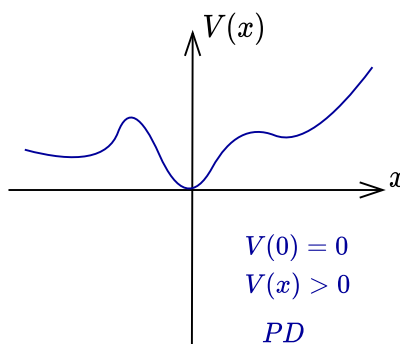
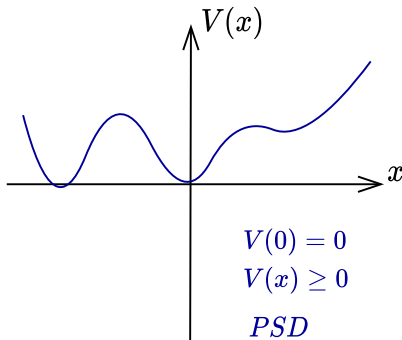
$PSD: (1) V(0) = 0$

$V(x) \geq 0, \forall x \in D - \{0\}$

$NSD: (1) V(0) = 0$

$V(x) \leq 0, \forall x \in D - \{0\}$

EX.



Recall

$\dot{x} = f(x)$

$x = 0$ 是平衡点 $\rightarrow f(x=0) = 0 : \dot{x}|_{x=0} = 0$

设一个 V

若 (1) $V: PD$

(2) $\dot{V}: NSD$

$x = 0$ 是稳定点

若 (1) $V: PD$

(2) $\dot{V}: ND$

$x = 0$ 是渐进稳定点

例子

$\dot{x}_1 = ax_1 = f_1$

$\dot{x}_2 = bx_2 + \cos x_1 = f_2$

设

$V = x_1^2 + x_2^2$

$$\dot{V} = 2x_1 \dot{x}_1 + 2x_1 \dot{x}_1 = 2x_1 ax_1 + 2x_2 (bx_2 + \cos x_1) = 2ax_1^2 + 2bx_2^2 + 2x_2 \cos x_1$$

$$= \begin{bmatrix} 2x_1 & 2x_2 \end{bmatrix} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} \frac{\partial V}{\partial x_1} & \frac{\partial V}{\partial x_2} \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \nabla V \cdot f(x) = Lf V(x)$$

Lie Derivative

Ex. pendulum without friction

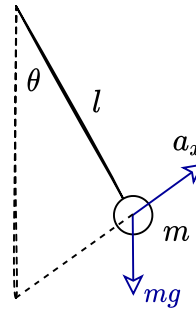
$$l\ddot{\theta} + g \sin \theta = 0$$

线性化 $\sin \theta \rightarrow \theta, \theta \approx 0$

$$\ddot{\theta} + \frac{g}{l} \sin \theta = 0$$

$$\ddot{\theta} + \frac{g}{l} \theta = 0$$

$$\text{周期: } T = 2\pi \sqrt{\frac{g}{l}}$$



$$\Sigma F_x = ma_x = l\ddot{\theta}$$

$$-mg \sin \theta = ml\ddot{\theta}$$

设 $x_1 = \theta, x_2 = \dot{x}_1 = \dot{\theta}$

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -\frac{g}{l} \sin x_1 \end{cases}$$



Find "V"

Energy

$$E = K + P = V$$

kinetic potential

$$V = \frac{1}{2}mv^2 + mgh$$

$$v = l\dot{\theta} = lx_2$$

$$h = l(1 - \cos \theta) = l(1 - \cos x_1)$$

$$V = \frac{1}{2}m(lx_2)^2 + mgl(1 - \cos x_1) = \frac{1}{2}ml^2x_2^2 + mgl(1 - \cos x_1) \quad x_1 \in (-2\pi, 2\pi)$$

分析

① $V(0) = 0$

$V(x)$ PD

② $V(x) > 0$

$$\begin{aligned} \dot{V}(x) = \nabla V f &= \begin{bmatrix} \frac{\partial V}{\partial x_1} & \frac{\partial V}{\partial x_2} \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} mgl \sin x_1 & ml^2 x_2 \end{bmatrix} \begin{bmatrix} x_2 \\ -\frac{g}{l} \sin x_1 \end{bmatrix} \\ &= mglx_2 \sin x_1 - mglx_2 \sin x_1 = 0 \end{aligned}$$

③ $\dot{V}(x) = 0 \leq 0$

$\dot{V}(x)$ NSD

$V(x)$ PD

$\dot{V}(x)$ NSD



Stable

① x_1, x_2 有界

② 不代表 $x_1, x_2 \rightarrow 0$

V : 能量

$\dot{V}$: 能量随时间的变化 $\dot{V} = 0 \rightarrow$ 能量不变

+ Friction 摩擦

$$ml\ddot{\theta} = -mg\sin\theta - \underbrace{kl\dot{\theta}}_{\text{friction}}$$

$$x_1 = \theta, \quad x_2 = \dot{x}_1 = \dot{\theta}$$

$\dot{x}_1 = x_2$	f_1
$\dot{x}_2 = -\frac{g}{l}\sin x_1 - \frac{k}{m}x_2$	f_2

$$V = \frac{1}{2}ml^2x_2^2 + mgl(1 - \cos x_1) \quad x_1 \in (-2\pi, 2\pi)$$

$$\dot{V}(x) = \nabla V f = \begin{bmatrix} \frac{\partial V}{\partial x_1} & \frac{\partial V}{\partial x_2} \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} mgl\sin x_1 & ml^2x_2 \end{bmatrix} \begin{bmatrix} x_2 \\ -\frac{g}{l}\sin x_1 - \frac{k}{m}x_2 \end{bmatrix}$$

$$= mglx_2\sin x_1 - mglx_2\sin x_1 - kl^2x_2^2$$

$$= -kl^2x_2^2$$

$ND?$ 只有 $x = [0, 0]$ $\dot{V} = 0$

NSD $x = [x_1, 0]$ $\dot{V} = 0$ (x_1 可以是任意值)

有摩擦, 物理 $t \rightarrow \infty \quad x_1 \ x_2 \rightarrow 0$

在这里, 数学上只判断了系统稳定, 并没有判断它渐进稳定

Advanced 控制理论 nonlinear 非线性

Invariance Principle -- LaSalle's Theorem

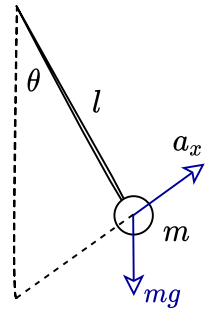
Motivation: pendulum with Friction

$$\ddot{\theta} = -\frac{g}{l}\sin\theta - \frac{k}{m}\dot{\theta}$$

$$x_1 = \theta, \quad x_2 = \dot{x}_1 = \dot{\theta}$$

$$\dot{x}_1 = x_2 \quad \text{①}$$

$$\dot{x}_2 = -\frac{g}{l}\sin x_1 - \frac{k}{m}x_2 \quad \text{②}$$



let $V = T + E = V = \frac{1}{2}ml^2\dot{x}_2^2 + mgl(1 - \cos x_1)$ **PD** ③

$$\dot{V}(x) = \nabla V f = \begin{bmatrix} \frac{\partial V}{\partial x_1} & \frac{\partial V}{\partial x_2} \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} mgl \sin x_1 & ml^2 x_2 \end{bmatrix} \begin{bmatrix} x_2 \\ -\frac{g}{l}\sin x_1 - \frac{k}{m}x_2 \end{bmatrix}$$

$$= -kl^2 x_2^2 \quad \text{④}$$

NSD

$$\dot{V} = 0$$

$$x = [x_1, 0]$$

$V : PD$
 $\dot{V} : NSD$ $\Rightarrow$ Stable but not AS

物理告诉我们 $t \rightarrow \infty$
 $x_1 \ x_2 \rightarrow 0$

引入新的理论, 扩大 Liapunov 稳定性的判据

system (1): $\dot{x} = f(x)$

$$R = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \begin{matrix} x_1 \in (-\pi, \pi) \\ x_2 \in (-a, a) \end{matrix} \quad a \in \mathbb{R}^+$$

Theorem S : the equilibrium point $x = 0$ of the autonomous system (1) is asymptotically stable if there exists a function $V(x)$ satisfying.

- (i) $V(x)$ positive definite $\forall x \in D$, where we assume that $0 \in D$ **PD**
- (ii) $\dot{V}(x)$ is negative semi definite in a bounded region $R \subset D$ **NSD**
- (iii) $\dot{V}(x)$ does not vanish identically along any trajectory in R , other than the null solution $x = 0$. $\neq 0$

除了零点以外都不等于0, 只有零点为0

检验第三点

$$\dot{V} = 0$$

$$\text{④} \rightarrow x_2 = 0, \forall t \quad \dot{x}_2 = 0$$

$$\text{②} \rightarrow 0 = -\frac{g}{l}\sin x_1$$

$$\Rightarrow x_1 = 0$$

$$\Rightarrow \dot{V} = 0 \text{ only if } x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

满足第三条, asymptotically

LaSalle's Theorem可以参考: <https://zhuanlan.zhihu.com/p/31925435>

Review pendulum without Friction

可以回顾一下没有摩擦的情况, 是不满足的

E.g. $\dot{x}_1 = x_2$ ----- ①

$\dot{x}_2 = -x_1 - x_2 - (x_1 + x_2)^2 x_2$ ②

$V = x_1^2 + x_2^2$ *PD*

$$\dot{V}(x) = \begin{bmatrix} \frac{\partial V}{\partial x_1} & \frac{\partial V}{\partial x_2} \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} 2x_1 & 2x_2 \end{bmatrix} \begin{bmatrix} x_2 \\ -x_1 - x_2 - (x_1 + x_2)^2 x_2 \end{bmatrix}$$

$$= 2x_1 x_2 - 2x_1 x_2 - 2x_2^2 - 2x_2^2 (x_1 + x_2)^2$$

$$= -2x_2^2 (1 + (x_1 + x_2)^2)$$

NSD

$x = [x_1, 0]$

$\dot{V}(x) = 0$

只需要检测 $\dot{V} = 0$ 是不是只有 $x = [0, 0]$ 满足。

$\dot{V} = 0 \quad \Rightarrow \quad x_2 = 0, \forall t \quad \dot{x}_2 = 0$

$$0 = -2x_2^2 (1 + (x_1 + x_2)^2)$$

$$x_1 = 0$$

$\dot{V} = 0$, 必须是 $x_1 = 0, x_2 = 0$ $\Rightarrow$ *Asymptotically stable*

Advanced 控制理论 nonlinear 非线性

Basic Feedback Stabilization

考虑一个系统

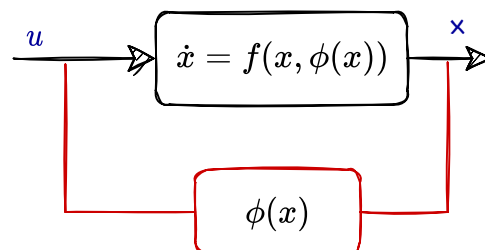
$$\dot{x} = f(x, u)$$

假设 u 是 x 的函数: state feedback

$$u = \phi(x)$$

$$\dot{x} = f(x, \phi(x))$$

研究对象



E.X1

$$\dot{x} = f(x, u) = ax^2 + u$$

$$f(0, 0) = 0$$



平衡点在原点(origin)

$$u = \phi(x)$$

若希望"0"是渐进稳定平衡点 ← Stability of origin

直觉告诉我们

$$u = -ax^2 - x \text{ then } \dot{x} = -x$$

消除了非线性项

提供了稳定

$$x(t) = Ce^{-t}$$

$$t \rightarrow \infty \quad x(t) \rightarrow 0$$

观察

- ① "a" 必须知道
② 这个方法很直接, 简单粗暴, 但不一定最佳

Feedback Linearization

反馈系统线性化, 非线性的系统, 通过输入, 线性化

E.X2

$$\dot{x} = x^2 - x^3 + u$$

$$u_1 = -x^2 + x^3 - x$$

代替方法:

Lyapunov Direct Method

回忆

$$\dot{x} = f(x) \quad \dot{x} = 0, x = 0$$

$$V: \quad V(0) = 0$$

$$V(x) : PD$$

$$\dot{V}(x) : ND$$

 $x = 0$ 是 Asymptotically stable

$$\dot{x} = x^2 - x^3 + u$$

$$\text{let } V = \frac{1}{2}x^2$$

$$V(0) = 0$$

$$V(x) : PD \quad V(x) > 0 \text{ in } x \neq 0$$

$$\dot{V} = \frac{\partial V}{\partial x} \frac{dx}{dt} = x\dot{x} = x(x^2 - x^3 + u) = \overset{\text{消除}}{x^3} - \overset{\text{not ND}}{x^4} + \overset{\text{ND}}{xu}$$

$$\text{let } u_2 = -x^2 - x \Rightarrow \dot{V} = -x^4 - x^2 \quad \text{ND}$$

$$u_3 = -x^2 \Rightarrow \dot{V} = -x^4 \quad \text{ND}$$

$$\dot{x} = x^2 - x^3 + u$$

$$\textcircled{1} \quad u_1 = -x^2 + x^3 - x$$

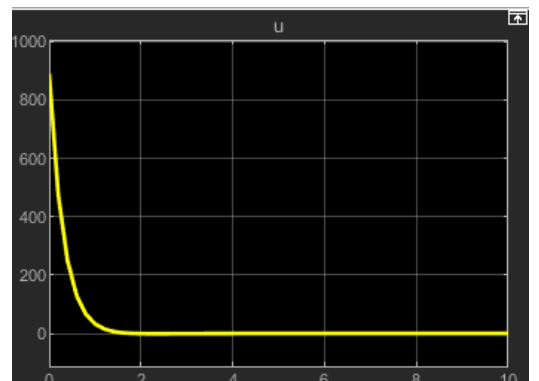
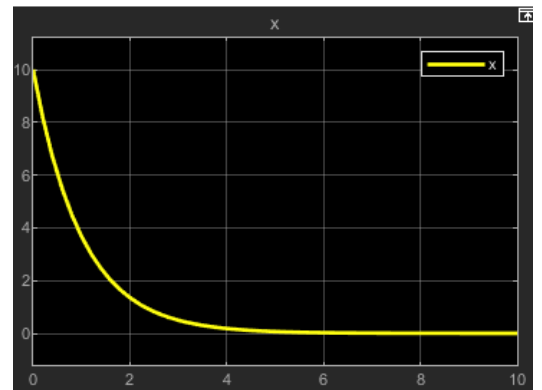
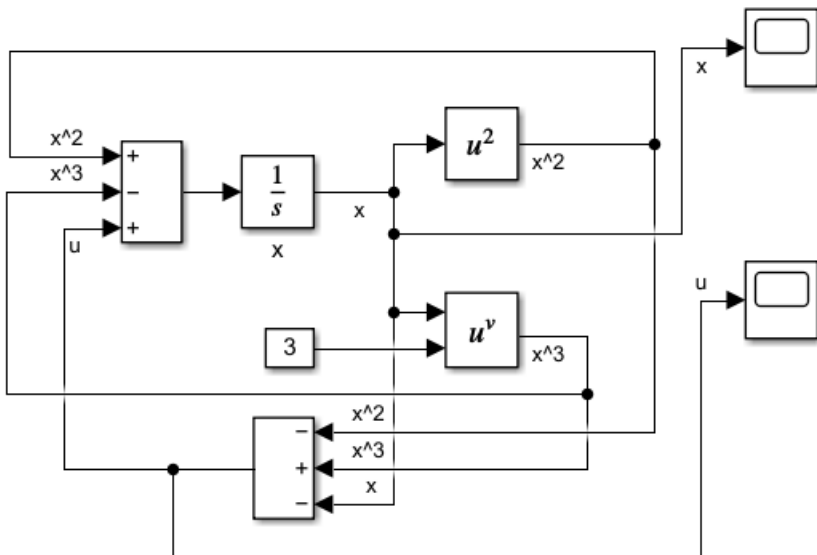
$$\textcircled{2} \quad u_2 = -x^2 - x$$

$$\textcircled{3} \quad u_3 = -x^2$$

simulink, 仿真对比这三种的结果

第一种

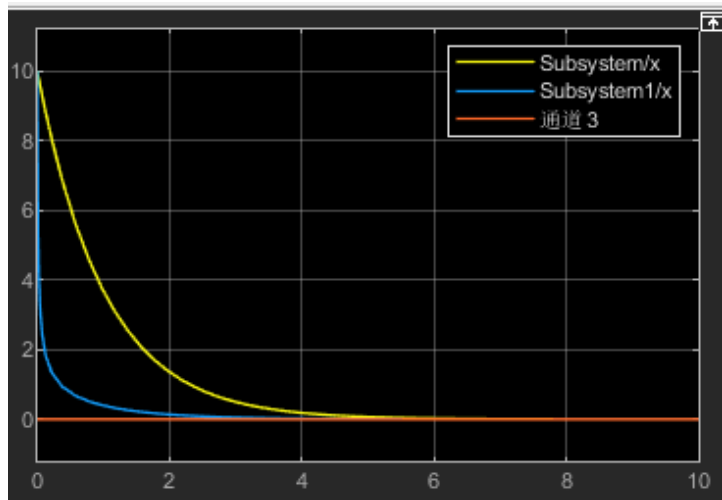
初始值为10



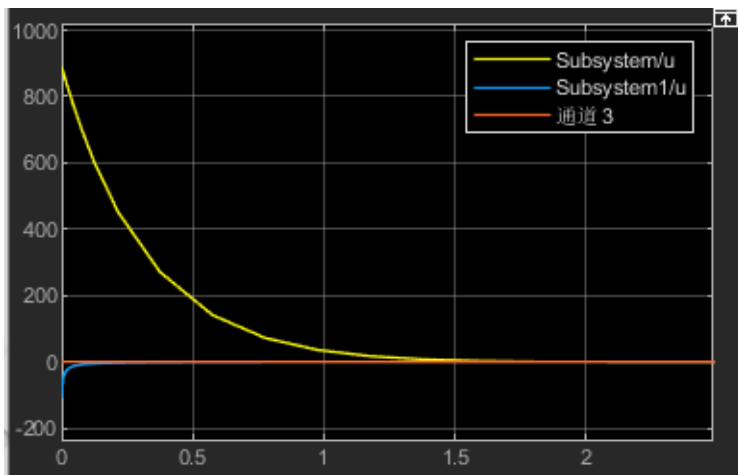
输入很高：因为有 x^3 项。
这也是我们要极力避免这种简单粗暴方法的原因。
往往会引入一个很大项，导致输入很大

①和②的对比

x:

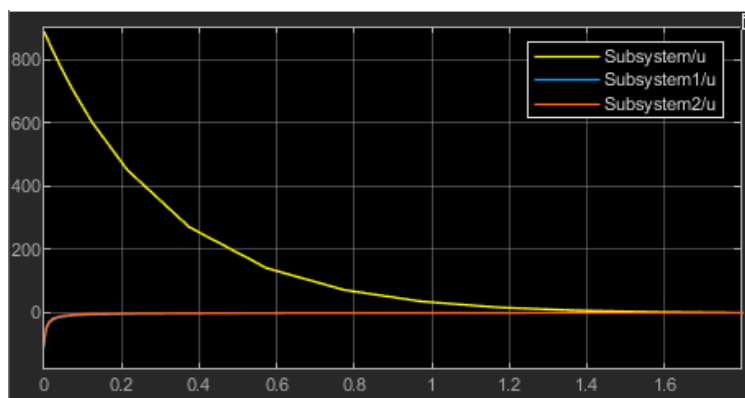
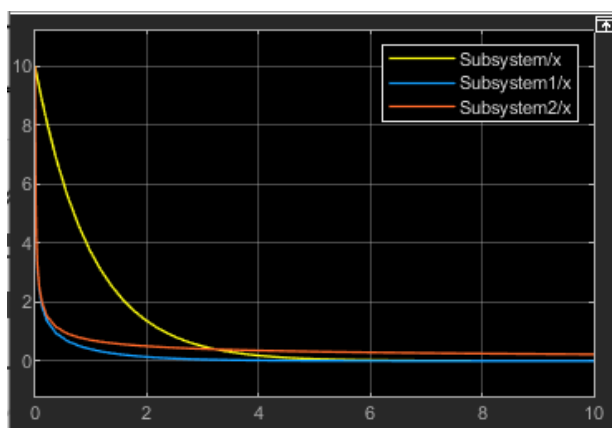


u



可以看到，②比①收敛的速度更快。输入u，②比①也小很多

①②③的对比



②③刚开始比①收敛的速度更快，但是③到后面越来越慢。

输入u，②③都比①小很多

$$\textcircled{1} \quad \dot{x} = -x \quad \Rightarrow \quad x = C_1 e^{-t}$$

$$\textcircled{2} \quad \dot{x} = -x^3 - x \quad \Rightarrow \quad x = \frac{1}{\sqrt{e^{C_1+2t}} - 1}$$

$$\textcircled{3} \quad \dot{x} = -x^3 \quad \Rightarrow \quad x = \frac{1}{\sqrt{C_1 + 2t}}$$

可以看到，收敛速度来说，①②有指数项，比③要快很多

验证了，在设计u2控制器时，引入的-x项，对系统的收敛速度有很积极的作用

Advanced 控制理论 nonlinear 非线性

Nonlinear Backstepping Control

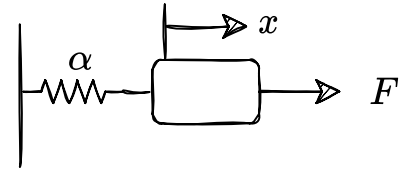
重要! 重要! 重要!

前方高能, 大量数学计算

拿出纸笔, let's Rock & Roll!

EX.

令: $F = u$ 输入 Input
 $x_1 = x$ 位移 Displacement
 $x_2 = \dot{x}$ 速度 velocity
 $x_1 \rightarrow x_{1d}$ Desired 规定轨迹



$$m\ddot{x} + \alpha x^3 = F$$

目标: 改变F使得滑块按指定轨迹移动

整理 $\Rightarrow \dot{x}_1 = \dot{x} = x_2$ (1)

$$\dot{x}_2 = \ddot{x} = -\frac{\alpha}{m}x_1^3 + \frac{1}{m}u$$
 (2)

通过改变u, 控制x2, 进一步控制x1
 即控制对滑块的力->速度->位置

Chain of Integrator

EX 空调
油门

引入 e: error

$$e = x_{1d} - x_1$$
 (3) 目标 $e \rightarrow 0$

代入(2)

$$\dot{e} = \dot{x}_{1d} - \dot{x}_1 = \dot{x}_{1d} - x_2$$
 (4)

找到Lyapunov: $V(e)$

$$V(e) : PD \quad \dot{V}(e) : ND \quad e \rightarrow 0$$

$$\text{设: } V_1 = \frac{1}{2}e^2$$
 (5) PD $V(0) = 0 \quad V(e \neq 0) > 0$

$$\dot{V}_1 = \frac{\partial V_1}{\partial e} \frac{de}{dt} = e\dot{e} = e(\dot{x}_{1d} - x_2)$$
 (6) $= -k_1 e$

希望 $\dot{V}_1$ ND, $x_{2d} = \dot{x}_{1d} + k_1 e$ (7)
 Desired

$$\begin{matrix} k_1 > 0 \\ \text{Gain} \\ \dot{V}_1 = -k_1 e^2 \\ ND \end{matrix}$$

新的目标 $x_2 \rightarrow x_{2d}$

$$\text{设 } \delta = x_{2d} - x_2$$
 (8) 目标 $\delta \rightarrow 0$

把(8)代入(6) $\dot{V}_1 = e(\dot{x}_{1d} - (x_{2d} - \delta))$ 代入(7)

$$\dot{x}_1 = x_2$$
 (1)

$$\dot{x}_2 = -\frac{\alpha}{m}x_1^3 + \frac{1}{m}u$$
 (2)

$$e = x_{1d} - x_1$$
 (3)

$$\dot{e} = \dot{x}_{1d} - x_2$$
 (4)

$$V_1 = \frac{1}{2}e^2$$
 (5)

$$\dot{V}_1 = e(\dot{x}_{1d} - x_2)$$
 (6)

$$x_{2d} = \dot{x}_{1d} + k_1 e$$
 (7)

$$\delta = x_{2d} - x_2$$
 (8)

$$\dot{V}_1 = -k_1 e^2 + e\delta \quad (9)$$

分析 $\delta = \dot{x}_{2d} - \dot{x}_2$ 代入(2)(7)

$$\dot{\delta} = \ddot{x}_{1d} + k_1 \dot{e} - \left(-\frac{\alpha}{m} x_1^3 + \frac{1}{m} u\right) \quad \text{代入(4)}$$

$$\dot{\delta} = \ddot{x}_{1d} + k_1 (\dot{x}_{1d} - x_2) + \frac{\alpha}{m} x_1^3 - \frac{1}{m} u \quad (10)$$

新的目标 $\delta, e \rightarrow 0$

$$\begin{array}{ll} V(e, \delta) & PD \\ \dot{V}(e, \delta) & ND \end{array}$$

设: $V_2 = V_1 + \frac{1}{2} \delta^2$ PD

$PD \quad PD$

$$\dot{V}_2 = \dot{V}_1 + \delta \dot{\delta} \quad \text{代入(9)}$$

$$\dot{V}_2 = -k_1 e^2 + e\delta + \delta \dot{\delta} = \underbrace{-k_1 e^2}_{ND} + \delta(e + \dot{\delta}) = -k_2 \delta$$

$$e + \dot{\delta} = -k_2 \delta \quad \text{代入(10)}$$

$k_2 > 0$
Gain

$$e + \ddot{x}_{1d} + k_1 (\dot{x}_{1d} - x_2) + \frac{\alpha}{m} x_1^3 - \frac{1}{m} u = -k_2 \delta$$

→ $u = me + m\ddot{x}_{1d} + mk_1 (\dot{x}_{1d} - x_2) + \alpha x_1^3 + mk_2 \delta \quad (11)$

$$\dot{V}_1 = -k_1 e^2 + e\delta \quad (9)$$

$$\dot{\delta} = \ddot{x}_{1d} + k_1 (\dot{x}_{1d} - x_2) + \frac{\alpha}{m} x_1^3 - \frac{1}{m} u \quad (10)$$

$$\begin{aligned} u = & me + m\ddot{x}_{1d} + mk_1 (\dot{x}_{1d} - x_2) \\ & + \alpha x_1^3 + mk_2 \delta \quad (11) \end{aligned}$$

检验:

(8)代入(4)

$$\dot{e} = \dot{x}_{1d} - (\dot{x}_{2d} - \delta) \quad \text{代入(7)}$$

$$\dot{e} = -k_1 e + \delta$$

(11)代入(10)

$$\dot{\delta} = \ddot{x}_{1d} + k_1 (\dot{x}_{1d} - x_2) + \frac{\alpha}{m} x_1^3 - e - \ddot{x}_{1d} - k_1 (\dot{x}_{1d} - x_2) - \frac{\alpha}{m} x_1^3 - k_2 \delta$$

$$\dot{\delta} = -e - k_2 \delta$$

线性的系统

$$\begin{bmatrix} \dot{e} \\ \dot{\delta} \end{bmatrix} = \begin{bmatrix} -k_1 & 1 \\ -1 & -k_2 \end{bmatrix} \begin{bmatrix} e \\ \delta \end{bmatrix}$$

Feedback
Linearization

$$\lambda_1 + \lambda_2 = \Lambda = -k_1 - k_2 < 0$$

$$\lambda_1 \cdot \lambda_2 = \Delta = k_1 k_2 + 1 > 0$$

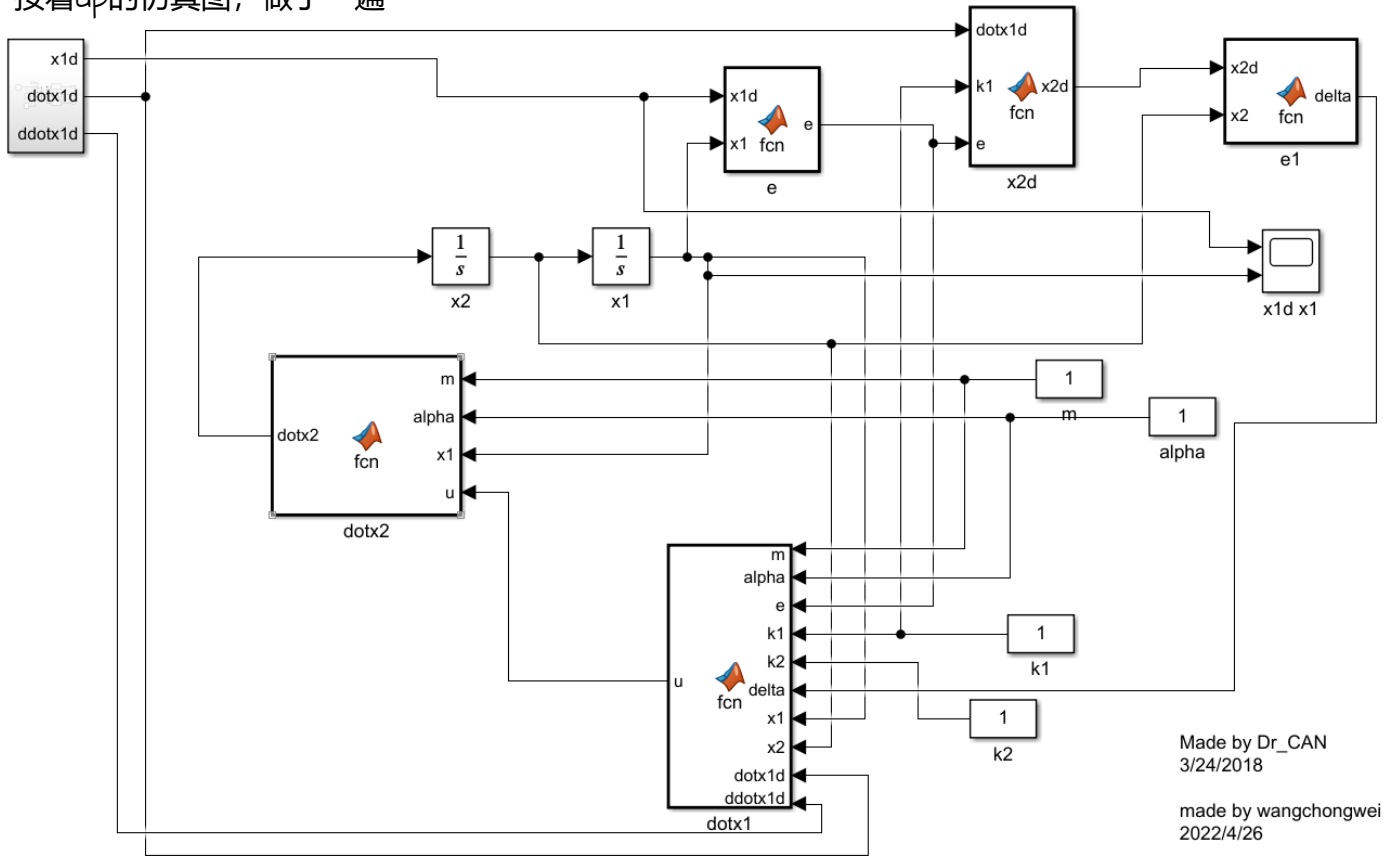
→ $\lambda_1, \lambda_2 < 0$

平衡点: $\begin{bmatrix} \dot{e} \\ \dot{\delta} \end{bmatrix} = 0 \rightarrow \begin{bmatrix} e \\ \delta \end{bmatrix} = 0$ Asymptotically Stable

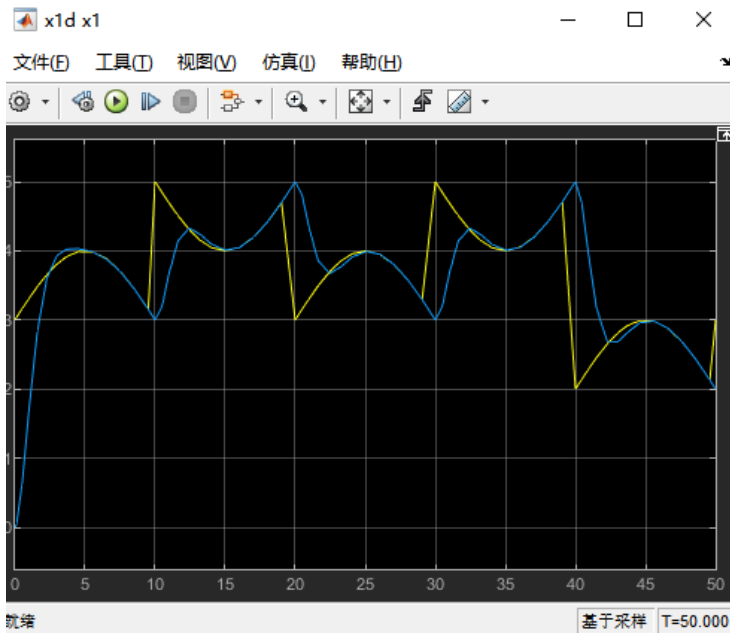
Rock

matlab simulink 仿真

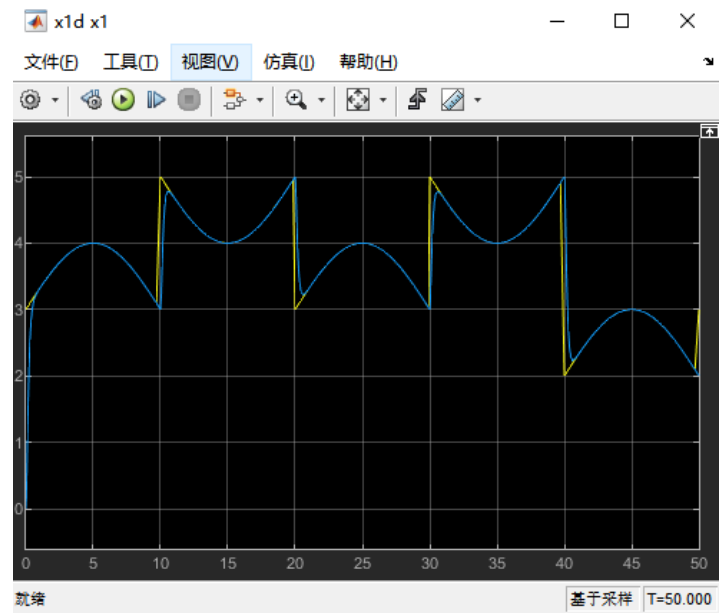
照着up的仿真图，做了一遍



$k1, k2$ 为的跟随效果



$k1, k2$ 为IO的跟随效果



Advanced 控制理论 nonlinear 非线性

Nonlinear Adaptive Controller 自适应控制器

$\dot{x} = ax^2 + u$ Tracking Problem $x \rightarrow x_d$ $e = x_d - x$ 目标 $e \rightarrow 0$

$\dot{e} = \dot{x}_d - \dot{x} = \dot{x}_d - ax^2 - u$ $\leftarrow \dot{e} = \dot{x}_d - ax^2 - \dot{x}_d + ax^2 - ke \Rightarrow \dot{e} = -ke$ Feedback Linearization

$V_{(e)}$ PD $\Rightarrow e \rightarrow 0$
 $\dot{V}_{(e)}$ ND

$V_{(e)} = \frac{1}{2}e^2$ PD
 $\dot{V}_{(e)} = \frac{\partial V}{\partial e} \dot{e} = e(\dot{x}_d - ax^2 - u)$
 $= e(\dot{x}_d - ax^2 - \dot{x}_d + ax^2 - ke) = -ke^2$ ND

$u = \dot{x}_d - ax^2 - ke$

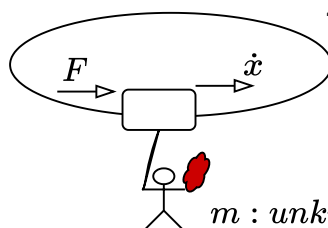
这个方式, 我们必须了解系统的所有参数

$\dot{x} = ax^2 + u$
 Exactly Known System

? if a is unknown, how?

Assumption: a 是常数, 缓慢变换

$\Rightarrow \dot{a} = 0$



2008年的火炬手, 假设不一定是李宁, 这个体重 m 不知道, 达到同样的效果

Adapt to this unknown parameter

$\dot{e} = \dot{x}_d - ax^2 - u$ (a unknown, $\dot{a} = 0$)

Estimate $\hat{a}$

Estimate error $\tilde{a} = a - \hat{a}$

$\dot{\tilde{a}} = \dot{a} - \dot{\hat{a}} \Rightarrow \dot{\tilde{a}} = -\dot{\hat{a}}$

Define Lyapunov function

$V_{(e,\tilde{a})} = \frac{1}{2}e^2 + \frac{1}{2}\tilde{a}^2$ PD

设计 $u = \dot{x}_d - \hat{a}x^2 + ke$

$\dot{V} = e\dot{e} + \tilde{a}\dot{\tilde{a}} = e(\dot{x}_d - ax^2 - u) - \tilde{a}\dot{\hat{a}}$

$= e(-\tilde{a}x^2 - ke) - \tilde{a}\dot{\hat{a}} = -ke^2 - \tilde{a}(ex^2 + \dot{\hat{a}})$
 ND $\Rightarrow \dot{\hat{a}} = -ex^2$

$\dot{V}_{(e,\tilde{a})} = -ke^2$

$\dot{V}_{(0,\tilde{a})} = 0$ NSD

引入新的引理

Lemma: if Lyapunov-like lemma

1) $V \geq 0$ PSD

2) $\dot{V} \leq -g(t)$, where $g(t) \geq 0$ $\dot{V} = -ke^2 \leq -(ke^2)$ $g(t) = ke^2$ PSD

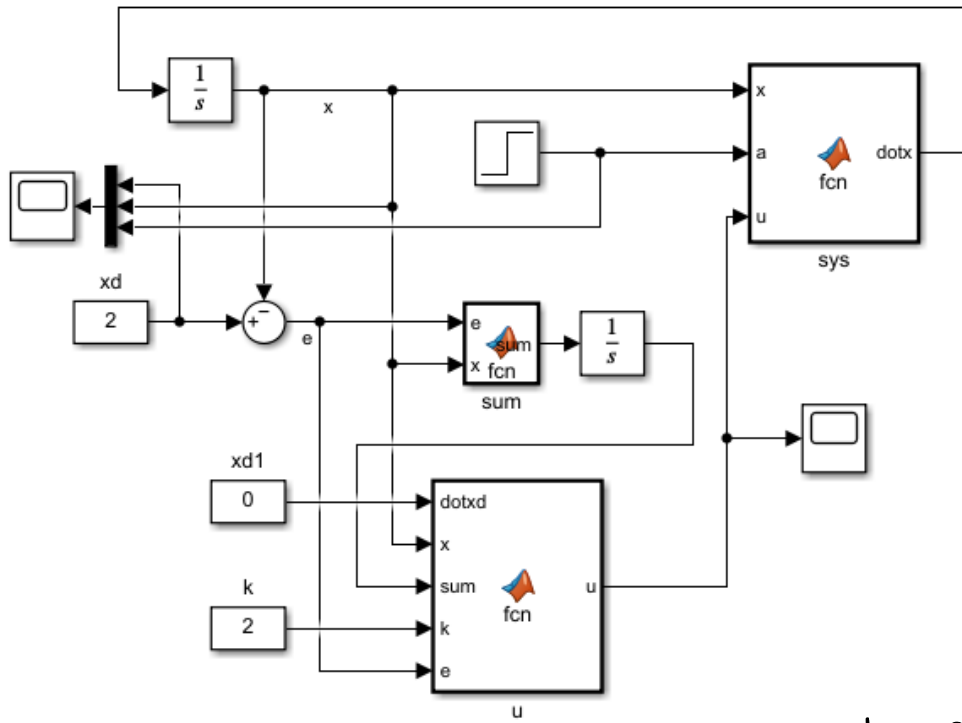
3) $\dot{g}(t) \in L_x$, if $\dot{g}(t)$ is bounded the $g(t)$ is uniformly continuous $\dot{g}(t) = 2ke\dot{e}$

then $\lim_{t \rightarrow \infty} g(t) = 0$ $\lim_{t \rightarrow \infty} ke^2 = 0 \Rightarrow e \rightarrow 0, t \rightarrow \infty$

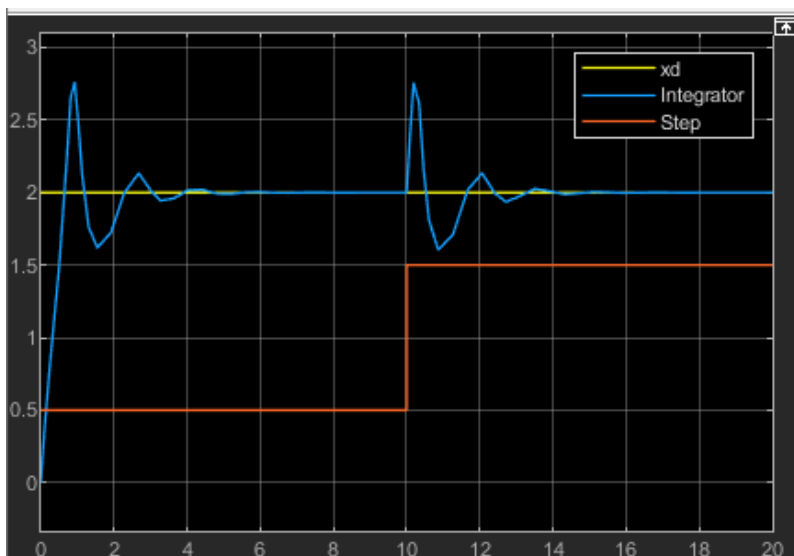
$$\dot{\hat{a}} = -ex^2 \quad \hat{a} = -\int_0^t ex^2 dt$$

$$u = \dot{x}_d + x^2 \int_0^t ex^2 dt + ke$$

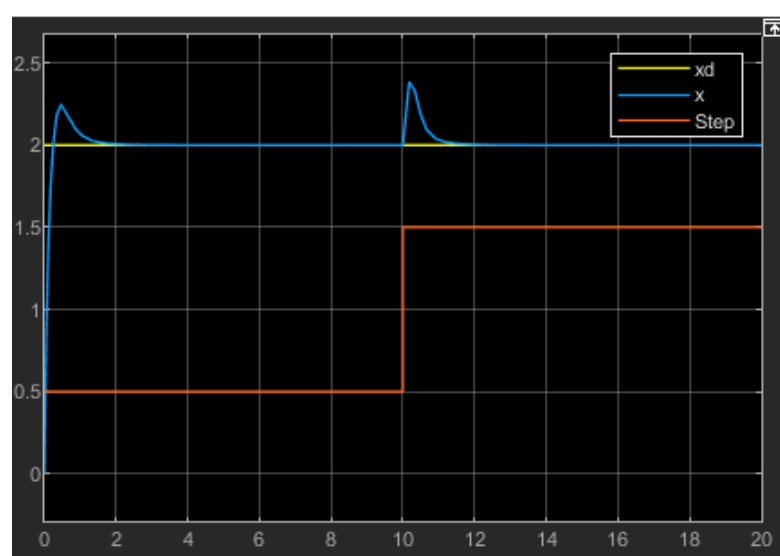
simulink:



$k = 2$



$k = 10$



Advanced 控制理论 nonlinear 非线性

Robust Controller 鲁棒控制

Sliding mode 滑模

Robust. 强壮



to achieve robust performance and/or stability in presence of bounded modeling error

存在一个有界的误差

- 空调
- 巡航

例子:

$$\dot{x} = ax^2 + u$$

Adaptive Controller 自适应

~~a is constant~~

$$u = \dot{x}_d + x^2 \int_0^t ex^2 dt + ke$$

$$x \rightarrow x_d$$

$$x_d - x = e \rightarrow 0$$

Robust

$$|a| \leq |\bar{a}|$$

a changing

如何设计控制器?

General

$$\dot{x} = f(x) + u$$

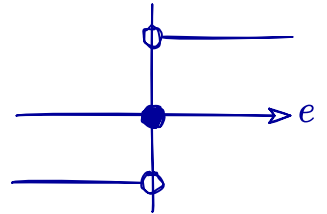
$$x \rightarrow x_d$$

$$\dot{e} = \dot{x}_d - \dot{x} = \dot{x}_d - f(x) - u$$

$$\text{condition: } |f(x)| < \rho(x)$$

$$\text{Let } u = \dot{x}_d + ke + \rho \frac{|e|}{e}$$

$$\frac{|e|}{e} = \text{sgn}(e) \begin{cases} 1 & e > 0 \\ 0 & e = 0 \\ -1 & e < 0 \end{cases}$$



Proof:

$$V_{(e)} = \frac{1}{2}e^2$$

$$\dot{V}_{(e)} = e\dot{e} = e(\dot{x}_d - f(x) - u) = e(\cancel{\dot{x}_d} - f(x) - \cancel{\dot{x}_d} - ke - \rho \frac{|e|}{e})$$

$$= -ke^2 - ef(x) - \rho|e|$$

$$\leq -ke^2 + \underbrace{|e|f(x)} - \rho|e|$$

$$\leq -ke^2 + |e|\rho - \rho|e|$$

$$\dot{V}_{(e)} \leq -ke^2$$

$$V_{(e)} = \frac{1}{2}e^2 \Rightarrow 2V_{(e)} = e^2$$

$$\dot{V}_{(e)} \leq -2kV_{(e)}$$

$$\dot{V}_{(e)} + 2kV_{(e)} \leq 0$$

微分方程不等式

Introduce

$$S_{(t)} > 0$$

$$\dot{V}_{(e)} + 2kV_{(e)} + S_{(t)} = 0$$

$$\dot{V}_{(e)} + 2kV_{(e)} = -S_{(t)}$$

微分方程

$$\exp(-2kt) = e^{-2kt}$$

避免和e混淆

$$V_{(t)} = V_{(0)} \exp(-2kt) - \underbrace{\exp(-2kt)}_{>0} \int_0^t \underbrace{\exp(2k\tau)}_{>0} \underbrace{S(\tau)}_{>0} d\tau$$

> 0

$$V_{(t)} \leq V_{(0)} \exp(-2kt)$$

$$\frac{1}{2} e_{(t)}^2 \leq \frac{1}{2} e_{(0)}^2 \exp(-2kt)$$

$$\sqrt{\frac{1}{2} e_{(t)}^2} \leq \sqrt{\frac{1}{2} e_{(0)}^2 \exp(-2kt)}$$

$$|e_{(t)}| \leq |e_{(0)}| \exp(-kt)$$

exponentially stable

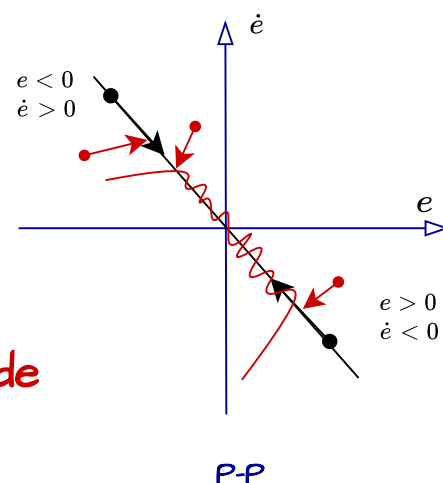
$$\dot{x} = f(x) + u$$

$$|f(x)| < \rho(x)$$

Let $u = \dot{x}_d + ke + \rho \frac{|e|}{e} \Rightarrow e = x_d - x \rightarrow 0, t \rightarrow \infty$

$$\dot{e} = \dot{x}_d - \dot{x} = \dot{x}_d - f(x) - u = -ke - f(x) - \rho(x) \frac{e}{|e|}$$

$$\dot{e} = -ke - \underbrace{f(x) - \rho(x) \frac{e}{|e|}}_{\text{controller}} \text{sgn}(e)$$



Sliding Mode

P-P

$$\dot{x} = ax^2 + u$$

$$|a| \leq |\bar{a}|$$

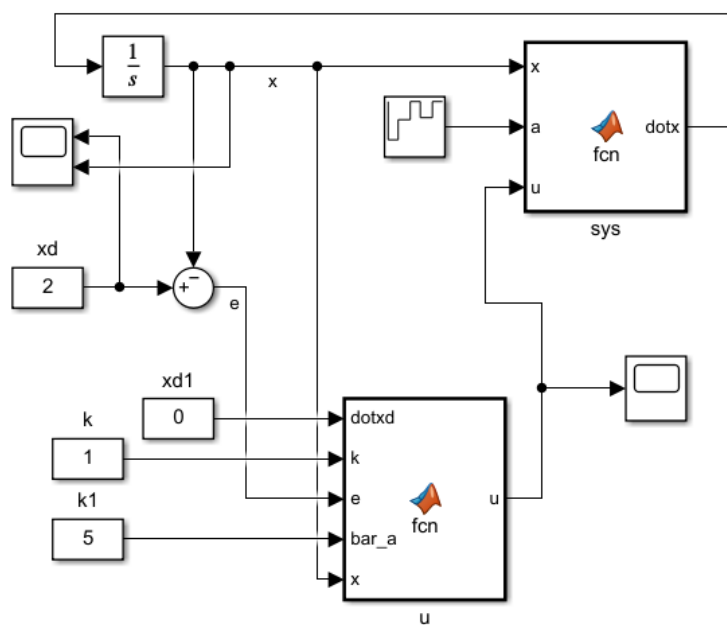
$$f(x)$$

$$|f(x)| = |a|x^2 \leq |\bar{a}|x^2 < |\bar{a}|(x^2 + 0.1)$$

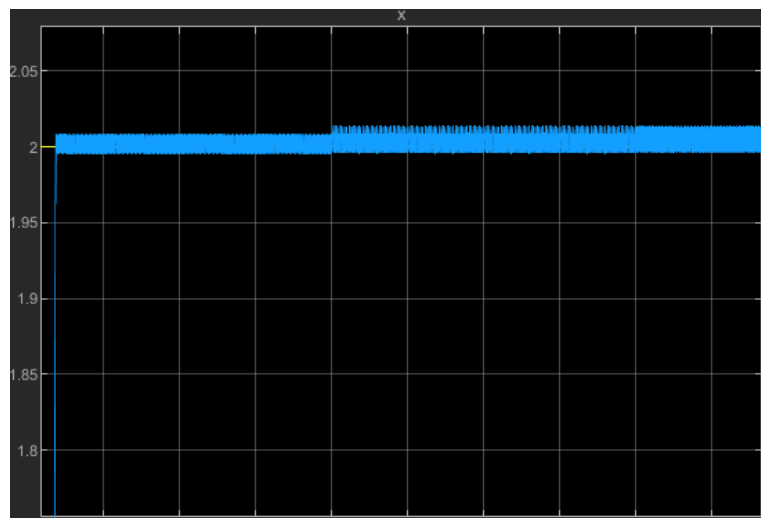
$\rho(x)$

$$u = ke + \dot{x}_d + |\bar{a}|(x^2 + 0.1) \frac{|e|}{e}$$

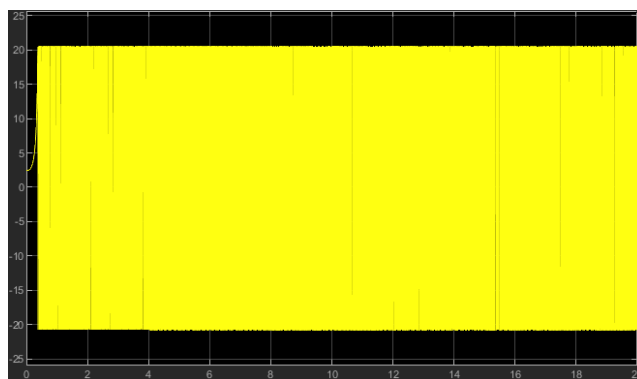
Simulink, 简单的仿真一下



x 和 x_d



我们看下输出 u , 输入很大



Advanced 控制理论 nonlinear 非线性 Robust Controller High gain and High Frequency

$$\dot{x} = f(x) + u \quad x \rightarrow x_d \quad e = x_d - x \rightarrow 0$$

$$f(x) : \text{uncertainty} \quad |f(x)| < \rho(x)$$

$$\dot{e} = \dot{x}_d - \dot{x} = \dot{x}_d - f(x) - u$$

$$\text{Let } u = ke + \dot{x}_d + u_{aux}$$

auxiliary

$$\text{Let } u = ke + \dot{x}_d + u_{aux}$$

回忆

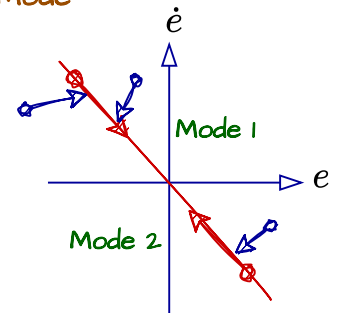
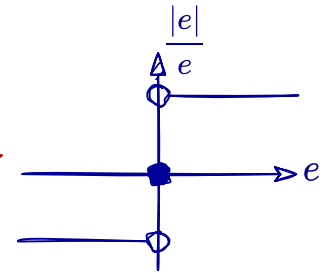
$$u_{aux} = \rho(x) \frac{e}{|e|} \quad \text{Sliding Mode}$$

代入

$$\dot{e} = -ke - f(x) - \rho(x) \frac{e}{|e|}$$

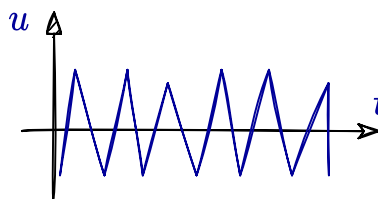
引到红色的线上

在两个mode中, 不断切换的



存在的问题: 输入来回切换

对于执行器来说是很大的挑战



$$\varepsilon > 0$$

$$\textcircled{2} \quad u_{aux2} = \frac{1}{\varepsilon} \rho^2 e$$

High Gain

用足够大的输入去抵消不确定性

$$\textcircled{3} \quad u_{aux3} = \frac{\rho^2 e}{\rho|e| + \varepsilon}$$

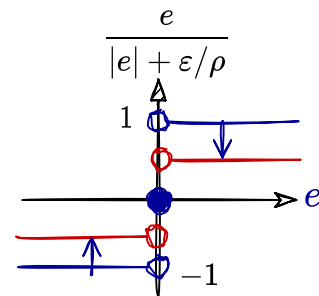
High Frequency

切换没那么剧烈, 平和一些

$$\varepsilon > 0$$

$$\frac{\rho^2 e}{\rho|e| + \varepsilon} \begin{cases} \varepsilon = 0, \frac{\rho^2 e}{\rho|e| + \varepsilon} = \rho \frac{e}{|e|} \\ \varepsilon \neq 0, \frac{\rho^2 e}{\rho|e| + \varepsilon} = \rho \frac{e}{|e| + \varepsilon/\rho} \end{cases} \quad \text{Sliding}$$

绝对值小于1



Proof:

$$\textcircled{2} \quad u_{aux2} = \frac{1}{\varepsilon} \rho^2 e$$

$$\text{Let } V = \frac{1}{2} e^2$$

$$\dot{V} = e\dot{e} = e(\dot{x}_d - f(x) - \dot{x}_d - ke - \frac{1}{\varepsilon} \rho^2 e)$$

$$|f(x)| < \rho(x)$$

$$= -ef(x) - ke^2 - \frac{1}{\varepsilon} \rho^2 e^2$$

$$-ef(x) \leq |e||f(x)| \leq |e|\rho$$

$$\dot{V} \leq \rho|e| - ke^2 - \frac{1}{\varepsilon} \rho^2 |e|^2 = -ke^2 + \rho|e|(1 - \frac{1}{\varepsilon} \rho|e|)$$

$$\dot{V} \leq -ke^2 + \rho|e|(1 - \frac{1}{\varepsilon}\rho|e|)$$

Case 1: $\rho|e| > \varepsilon \Rightarrow \frac{1}{\varepsilon}\rho|e| > 1 \Rightarrow 1 - \frac{1}{\varepsilon}\rho|e| < 0 \Rightarrow \rho|e|(1 - \frac{1}{\varepsilon}\rho|e|) < 0$

$$\Rightarrow \dot{V} \leq -ke^2 + 0 \Rightarrow \dot{V} \leq -ke^2$$

exponentially stable

证明参考滑膜控制的笔记，一模一样的

Case 2: $\rho|e| \leq \varepsilon \Rightarrow \frac{1}{\varepsilon}\rho|e| \leq 1 \Rightarrow 1 \geq (1 - \frac{1}{\varepsilon}\rho|e|) \geq 0 \Rightarrow \varepsilon \geq \rho|e| > \rho|e|(1 - \frac{1}{\varepsilon}\rho|e|)$

$$\dot{V} \leq -ke^2 + \varepsilon$$

$$V_{(e)} = \frac{1}{2}e^2 \Rightarrow -ke^2 = -2kV$$

$$\dot{V} \leq -2kV + \varepsilon$$

微分方程不等式

Introduce $S_{(t)} > 0$

$$\dot{V} + 2kV = \varepsilon - S_{(t)} \quad \text{一阶常系数线性非齐次微分方程}$$

$$V_{(t)} = V_{(0)}\exp(-2kt) - \underbrace{\exp(-2kt)}_{>0} \int_0^t \underbrace{\exp(2k\tau)}_{>0} \underbrace{S(\tau)}_{>0} d\tau + \varepsilon \exp(-2kt) \int_0^t \underbrace{\exp(2k\tau)}_{>0} d\tau$$

$$\frac{1}{2}e^2 = V_{(t)} \leq V_{(0)}\exp(-2kt) - 0 + \frac{\varepsilon}{2k}(1 - \exp(-2kt))$$

$$= \frac{1}{2k} \int_0^t \exp(2k\tau) d(2k\tau)$$

$$= \frac{1}{2k} \exp(2k\tau) \Big|_0^t$$

$$= \frac{1}{2k} (\exp(2k\tau) - 1)$$

$$\sqrt{\frac{1}{2}e_{(t)}^2} \leq \sqrt{\frac{1}{2}e_{(0)}\exp(-2kt) + \frac{\varepsilon}{2k}(1 - \exp(-2kt))}$$

$$\varepsilon \exp(-2kt) \frac{1}{2k} (\exp(2k\tau) - 1)$$

$$= \frac{\varepsilon}{2k} (1 - \exp(-2kt))$$

$$|e_{(t)}| \leq \sqrt{|e_{(0)}| \underbrace{\exp(-2kt)}_0 + \frac{\varepsilon}{k} - \underbrace{\frac{\varepsilon}{k}\exp(-2kt)}_0}$$

$t \rightarrow \infty \quad \quad \quad 0 \quad \quad \quad 0$

$$|e_{(t)}| \leq \sqrt{\frac{\varepsilon}{k}}$$

Globally uniformly ultimately bounded

GUUB

$$\varepsilon \downarrow \quad e \rightarrow 0 \quad u_{aux2} = \frac{1}{\varepsilon} \rho^2 e \uparrow$$

权衡，多大的误差是能接受的，
且执行器没有那么大的负担

Proof: ③ $u_{aux3} = \frac{\rho^2 e}{\rho|e| + \varepsilon}$

Let $V = \frac{1}{2}e^2$

$$\dot{V} = e\dot{e} \leq -ke^2 + \rho|e| - e \frac{\rho^2 e}{\rho|e| + \varepsilon} = -ke^2 + \frac{\rho^2|e|^2 + \rho|e|\varepsilon - \rho^2 e^2}{\rho|e| + \varepsilon}$$

$$\dot{V} \leq -ke^2 + \varepsilon \frac{\rho|e|}{\rho|e| + \varepsilon} \quad 0 \leq \frac{\rho|e|}{\rho|e| + \varepsilon} \leq 1$$

$\dot{V} \leq -ke^2 + \varepsilon$

same as u_{aux2} $|e| \leq \sqrt{\frac{\varepsilon}{k}}$

$$\dot{x} = f(x) + u \quad x \rightarrow x_d \quad e = x_d - x \rightarrow 0$$

$$|f(x)| < \rho(x)$$

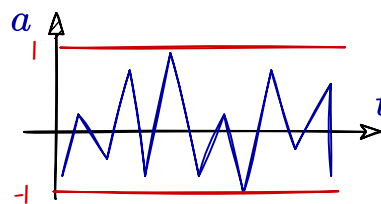
Let $u = ke + \dot{x}_d + u_{aux}$

$$u_{aux} = \begin{cases} \rho \frac{e}{|e|} & \text{sliding} \\ \frac{1}{\varepsilon} \rho^2 e & \text{high Gain} \\ \frac{\rho^2 e}{\rho|e| + \varepsilon} & \text{high frequency} \end{cases}$$

H. W **Simulink**

$$\dot{x} = ax^2 + u$$

$$a = f(t) \quad \text{随机} \quad |a| \leq 1$$



$$x \rightarrow x_d = 0$$

try $aux_{1,2,3}$

$$x_{(0)} = 0$$

$$\varepsilon = 1, 0.1, \dots$$

Advanced 控制理论 nonlinear 非线性 Robust Control

$$\dot{x} = f(x) + u$$

$$x \rightarrow x_d$$

$$|f(x)| < \rho(x)$$

$$u = ke + \dot{x}_d + u_{aux}$$

$$e = x_d - x \rightarrow 0$$

Sliding Mode 滑模

High Gain 高增益

High Frequency 高频

u_{aux}

$$\rho \frac{e}{|e|}$$

$$\frac{1}{\varepsilon} \rho^2 e, \varepsilon > 0$$

$$\frac{\rho^2 e}{\rho|e| + \varepsilon}, \varepsilon > 0$$

$$t \rightarrow \infty, e \rightarrow 0$$

$$|e| \leq \sqrt{\frac{\varepsilon}{k}}$$

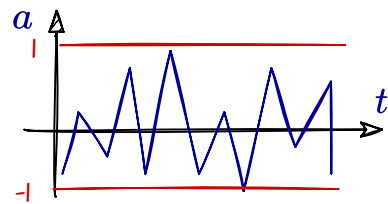
$$|e| \leq \sqrt{\frac{\varepsilon}{k}}$$

$$\varepsilon \downarrow e \downarrow u \uparrow$$

$$\dot{x} = ax^2 + u$$

$$|a| \leq 1 \quad \text{随机}$$

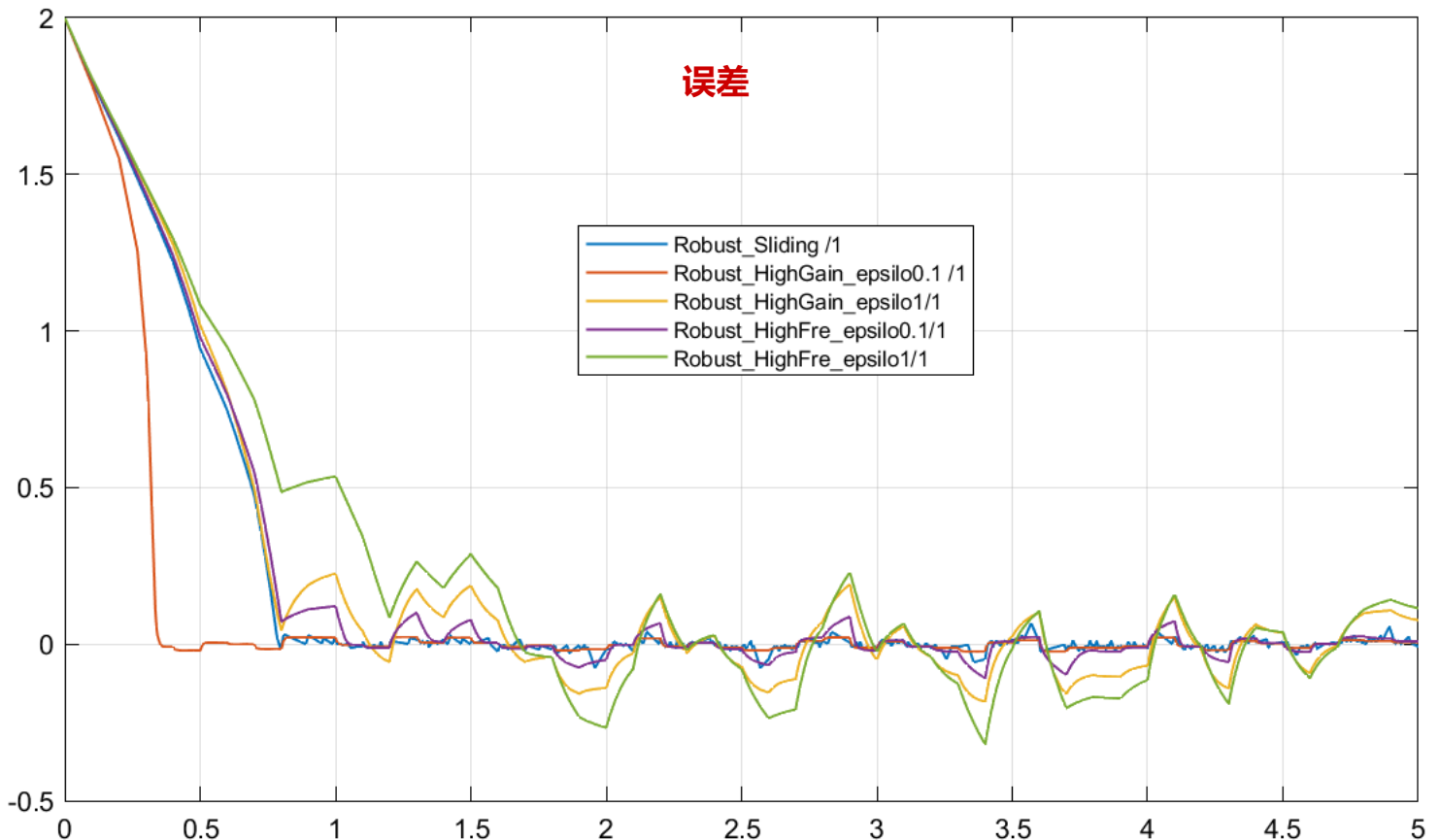
目标 $x \rightarrow x_d = 2 \Rightarrow \dot{x}_d = 0$

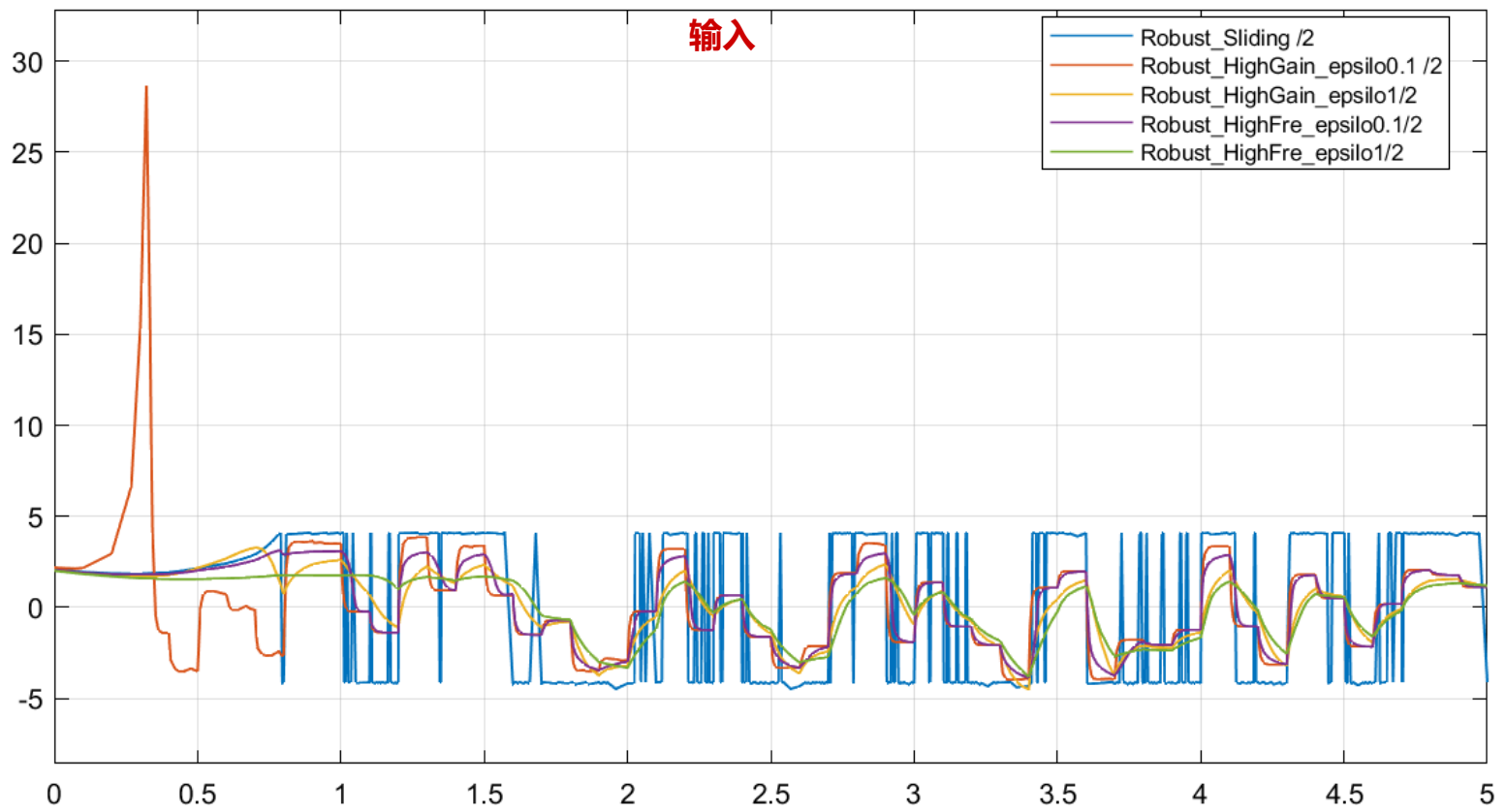


$$f(x) = ax^2 \leq |a||x|^2 \leq x^2 < x^2 + 0.1 = \rho(x)$$

$$u = ke + \dot{x}_d + u_{aux} = ke + u_{aux}$$

三种控制器的仿真





对比high-gain, $\epsilon=0.1$ 和 $\epsilon = 1$

对于误差, $\epsilon=0.1$ 比 $\epsilon = 1$, 小很多

$$|e| \leq \sqrt{\frac{\epsilon}{k}}$$

对于输入, $\epsilon=0.1$ 比 $\epsilon = 1$, 瞬态输入非常大, 稳态输入也很大

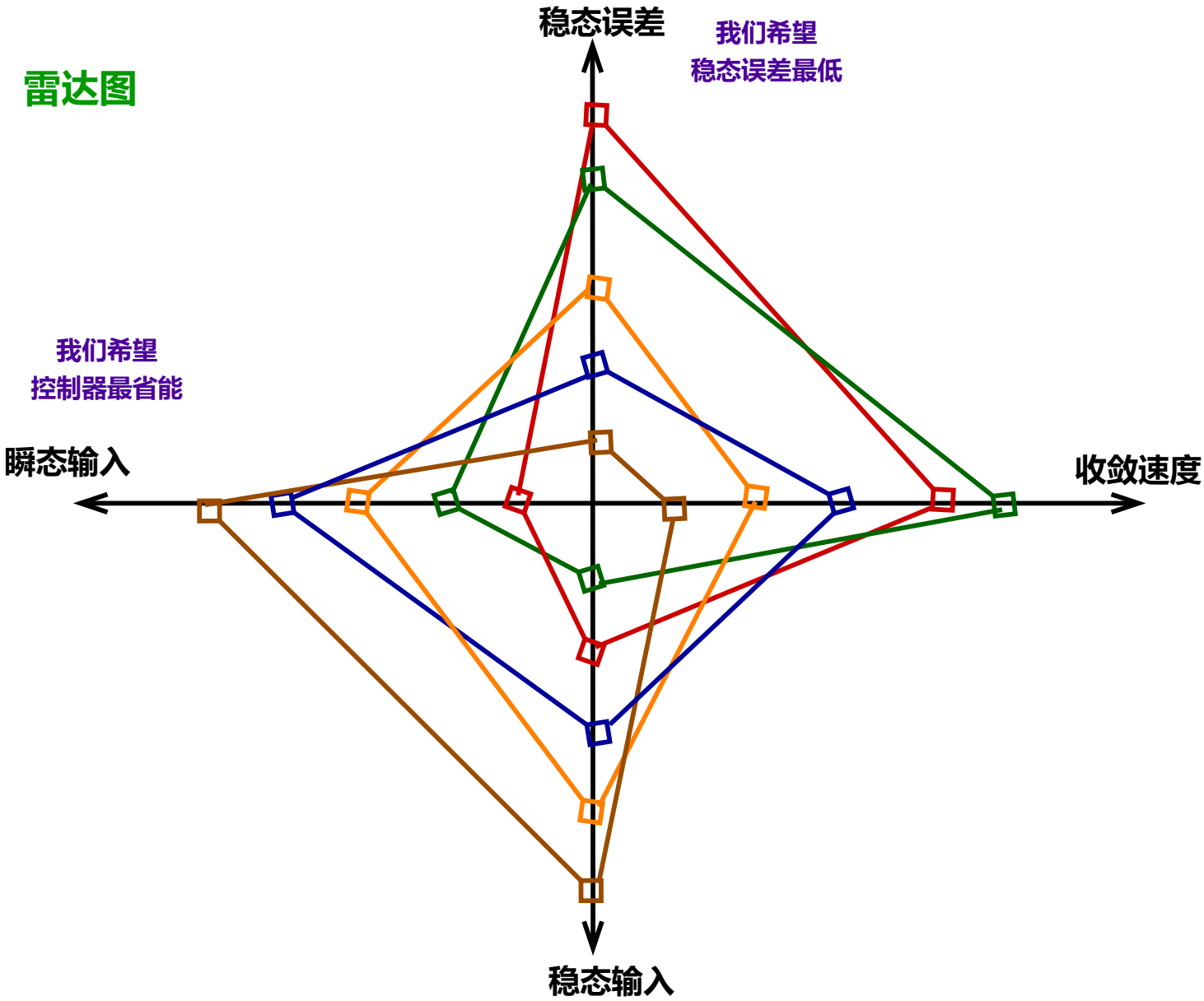
对于误差来说, 稳态误差越小, 分值越高

对于输入来说, 输入越小, 分值越高

Design of Experiments

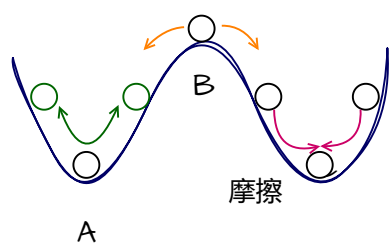
	Name	U_{aux}	Gain(k)	ϵ	稳态误差	收敛速度	瞬态输入	稳态输入
1	sliding mode	$\rho \frac{e}{ e }$	1	N/A	5	4	2	1
2	high gain	$\frac{1}{\epsilon} \rho^2 e$	1	0.1	4	5	1	2
4	high gain	$\frac{1}{\epsilon} \rho^2 e$	1	1	2	3	3	4
3	high freq	$\frac{\rho^2 e}{\rho e + \epsilon}$	1	0.1	3	2	4	3
5	high freq	$\frac{\rho^2 e}{\rho e + \epsilon}$	1	1	1	1	5	5

雷达图



Advanced 控制理论 nonlinear 非线性 串讲

Nonlinear System Lyapunov Stability



A和C的区别是有摩擦

1) A点, 球的偏离会一直摆下去

2) B点偏离稳定点后, 将远离

3) C点, 有摩擦, 偏离后最终回到C点

摩擦

$$\dot{x} = f(x)$$

$$\dot{x} = 0$$

$$\frac{dx}{dt} = 0$$

$$f(x_e) = 0$$

平衡点

不严谨的说: 稳定系统

离开平衡点后的反应随时间衰减, 或者至少不增加

A, C $\longrightarrow$ 稳定
B $\longrightarrow$ 不稳定

严谨的数学定义 $\epsilon - \delta$ $\forall$: for All: 对于任意给定 $\exists$: at least one: 存在一个 $\|\cdot\|$: norm, 范数Euclidean norm
欧几里得范数

$$\|x\| = \sqrt{x_1^2 + x_2^2 + x_3^2 + \cdots + x_n^2}$$

Defintion : Stability in the sense of Lyapunov

李雅普诺夫稳定性

The origin (equilibrium point at the origin) is stable in the sense of Lyapunov or simply stable if

$$\forall t_0, \forall \epsilon > 0, \exists \delta(t_0, \epsilon) : \|x(t_0)\| < \delta(t_0, \epsilon) \Rightarrow \forall t \geq t_0 \quad \|x(t)\| < \epsilon$$

Defintion : Asymptotic Stability 渐进稳定性

the origin is an asymptotically stable equilibrium point if it is stable and in addition:

$$\exists \delta(t_0) > 0 : \|x(t_0)\| < \delta(t_0) \Rightarrow \lim_{t \rightarrow \infty} \|x(t)\| = 0$$

这两个的区别在于

1) 一个有界

2) 一个随着时间的进行, x 的范数等于0

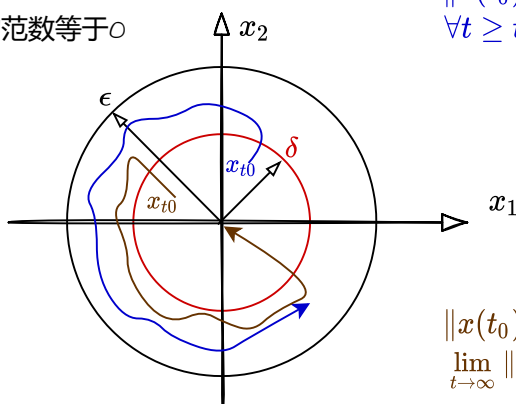
$$\|x(t_0)\| < \delta(t_0, \epsilon)$$

$$\forall t \geq t_0 \quad \|x(t)\| < \epsilon$$

最开始在小圆内随着时间的增加, 它会一直在大圆以内

2-D

$$\dot{x} = f(x_1, x_2)$$



$$\|x(t_0)\| < \delta(t_0, \epsilon)$$

$$\lim_{t \rightarrow \infty} \|x(t)\| = 0$$

最开始在小圆内随着时间的增加, 时间趋向无穷, 它会回到平衡点

Lyapunov Direct Method

$$\dot{x} = f(x)$$

$$x = 0 \text{ 是平衡点} \rightarrow f(x=0) = 0 : \dot{x}|_{x=0} = 0$$

设一个 V

若

(1) $V : PD$

(2) $\dot{V} : NSD$

 $x = 0$ 是稳定点

若

(1) $V : PD$

(2) $\dot{V} : ND$

 $x = 0$ 是渐进稳定点

$$PD: \quad (1) \quad V(0) = 0$$

$$V(x) > 0, \quad \forall x \in D - \{0\}$$

$$PSD: \quad (1) \quad V(0) = 0$$

$$V(x) \geq 0, \quad \forall x \in D - \{0\}$$

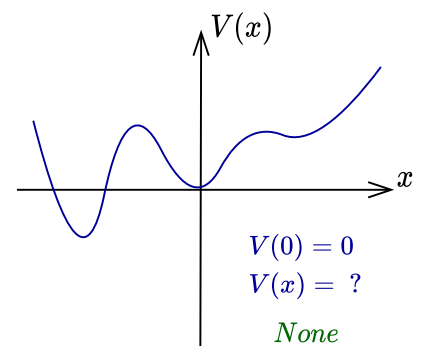
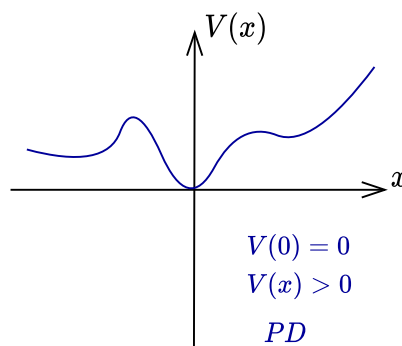
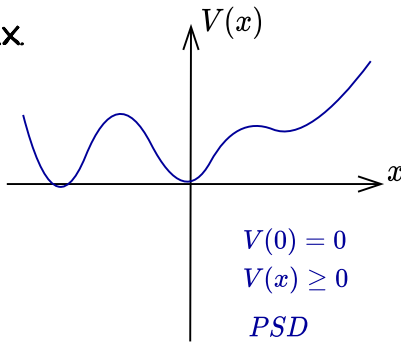
$$ND: \quad (1) \quad V(0) = 0$$

$$V(x) < 0, \quad \forall x \in D - \{0\}$$

$$NSD: \quad (1) \quad V(0) = 0$$

$$V(x) \leq 0, \quad \forall x \in D - \{0\}$$

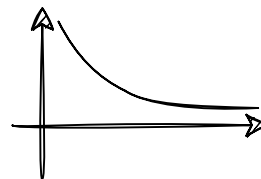
EX.



例子

$$\dot{x} = -x$$

$$x(t) = Ce^{-t}$$

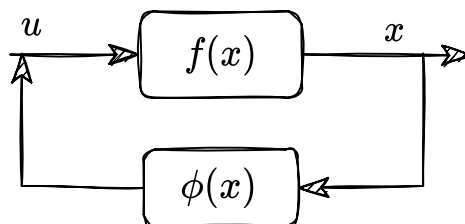


$$V(x) = \frac{1}{2}x^2 \quad PD$$

渐进稳定

$$\dot{V}(x) = x\dot{x} = x(-x) = -x^2 \quad ND$$

在控制系统中



$$\dot{x} = x^2 + u$$

$$x \rightarrow x_d$$

$$e = x_d - x$$

$$e \rightarrow 0$$

$$\dot{e} = \dot{x}_d - \dot{x} = \dot{x}_d - x^2 - u$$

希望是渐进稳定的, 平衡点在0点

$$V_{(e)} = \frac{1}{2}x^2 \quad PD$$

改变u, 使其ND **ctrl gain**

$$\dot{V}_{(e)} = e\dot{e} = e(\dot{x}_d - x^2 - u)$$

$$u = \dot{x}_d - x^2 + ke$$

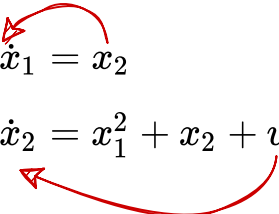
$$= e(-ke) = -ke^2 \quad ND$$

Feedback Linearization

$$\dot{e} = -ke$$

$$e = Ce^{-kt} \quad \text{改变k, 改变收敛速度}$$

二维:

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_1^2 + x_2 + u\end{aligned}$$


$$e = x_{1d} - x_1$$

$$x_1 \rightarrow x_{1d}$$

$$r = x_{2d} - x_2$$

$$x_2 \rightarrow x_{2d}$$

中间变量

Backstepping

$$V_{(e,r)} : PD$$

$$\dot{V}_{(e,r)} : ND$$

$$\dot{x} = a \sin x + x^2 + u$$

(a unknown, $\dot{a} = 0$)

Adaptive 自适应

Estimate $\hat{a}$

Estimate error $\tilde{a} = a - \hat{a}$

$$V_{(e,\tilde{a})} \quad PD$$

$$\dot{V} \quad ND$$

Robust.

$$\dot{x} = x^2 + f(x) + u$$

$$condition : |f(x)| < \rho(x)$$

$$u_{aux} = \begin{cases} \rho \frac{e}{|e|} & \text{Sliding} \\ \frac{1}{\varepsilon} \rho^2 e & \text{high Gain} \\ \frac{\rho^2 e}{\rho|e| + \varepsilon} & \text{high frequency} \end{cases}$$

$$V \quad PD$$

$$\dot{V} \quad ND$$

希望用一条线，串起来。这条线就是李雅普诺夫第二方法。